

brain-drain bay card: ASM1153E USB-SATA bridge, switched 12 V / 5 V, 22-pin SATA receptacle

bridge

ASM1153E USB 3 to SATA bridge, 30 MHz crystal, core switcher, AC coupling

File: sheets/bridge.kicad\_sch

bay-switch

Switched 12 V and 5 V for the drive: P-FET high-side switches with soft-start, PTC fuses

File: sheets/bay-switch.kicad\_sch

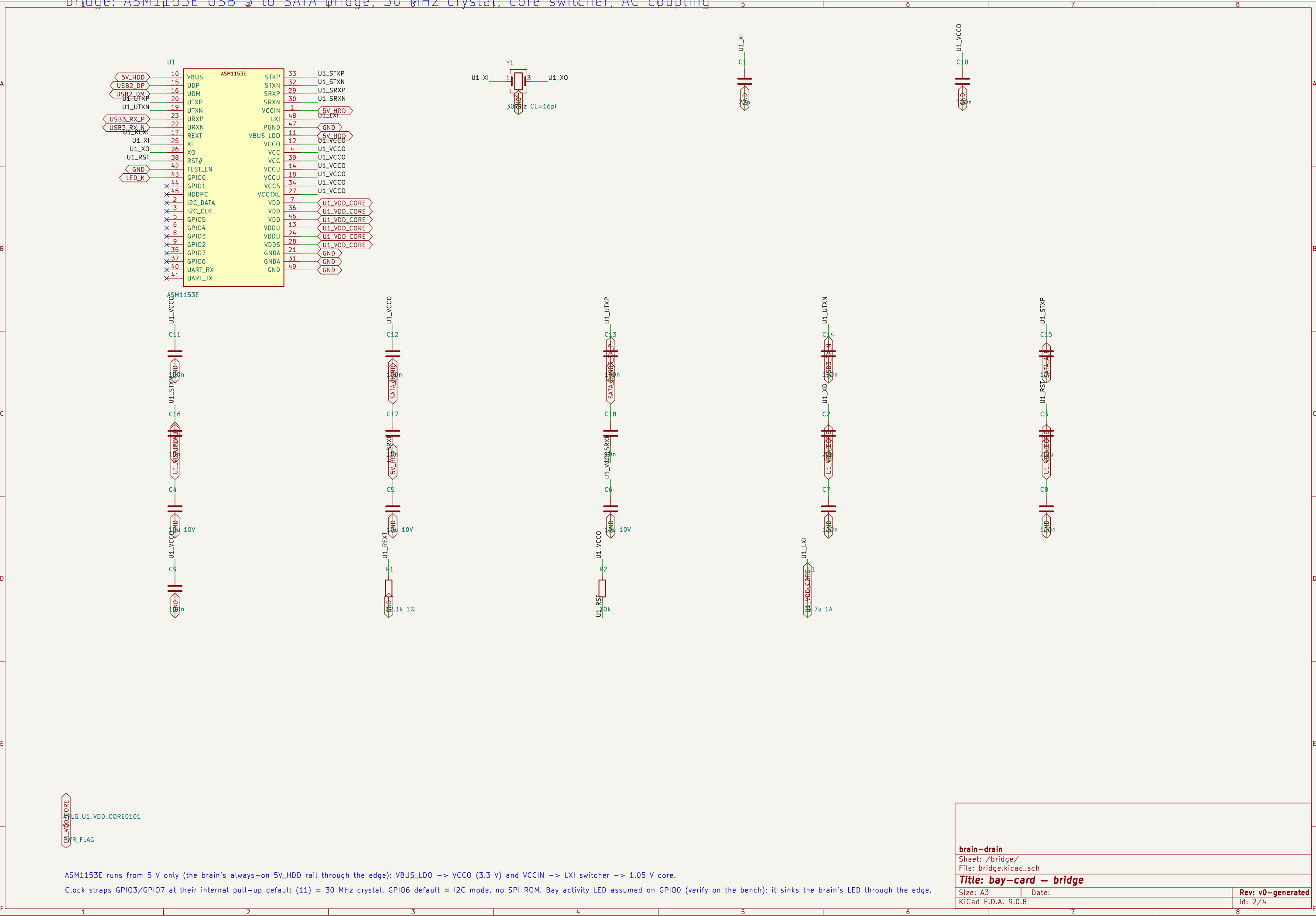
edge

Card edge to the brain (PCIe x1 fingers, bay-card pinout) and the SATA 22-pin receptacle

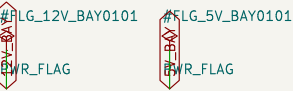
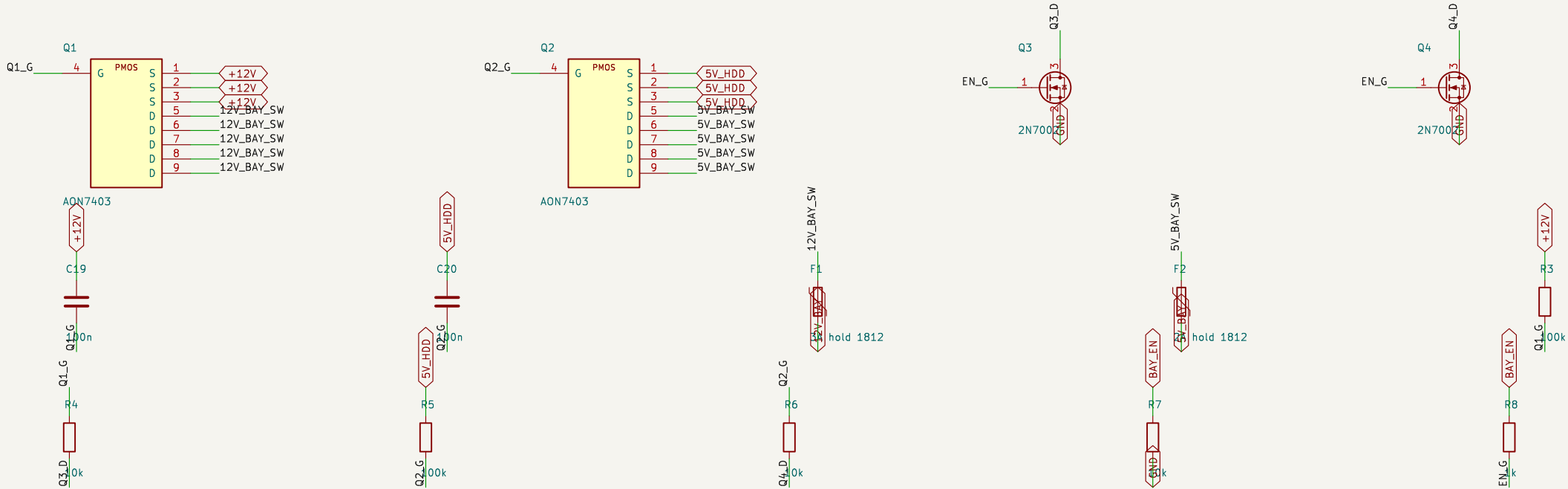
File: sheets/edge.kicad\_sch

|                                                                          |         |  |                   |
|--------------------------------------------------------------------------|---------|--|-------------------|
| brain-drain                                                              |         |  |                   |
| Sheet: /                                                                 |         |  |                   |
| File: bay-card.kicad_sch                                                 |         |  |                   |
| Title: brain-drain bay card: ASM1153E USB-SATA bridge, switched 12 V / 5 |         |  |                   |
| Size: A3                                                                 | Date:   |  | Rev: v0-generated |
| KiCad E.D.A. 9.0.8                                                       | Id: 1/4 |  |                   |

bridge: ASM1153E USB 3 to SATA bridge, 30 MHz crystal, core switcher, AC coupling

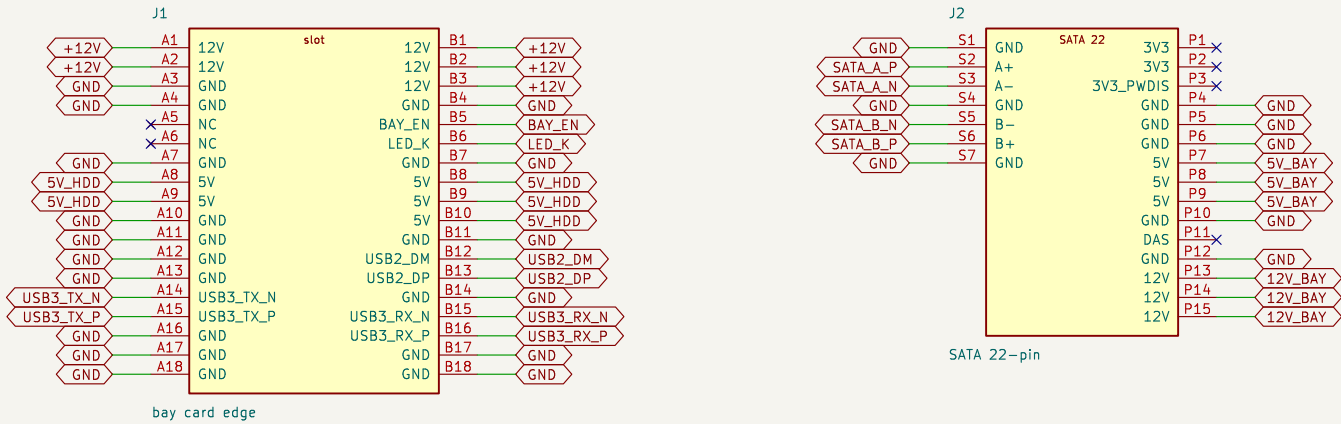


bay-switch: Switched 12 V and 5 V for the drive: P-FET high-side switches with soft-start, PTC fuses



P-FET high-side switch; 100k gate pull-up holds it off; the 2N7002 pulls the gate down through 10k with 100n gate-source, ~1 ms ramp so the drive's bulk caps do not trip the fuse. BAY\_EN pull-down keeps the bay off through boot

|                                                   |       |                   |
|---------------------------------------------------|-------|-------------------|
| brain-drain                                       |       |                   |
| Sheet: /bay-switch/<br>File: bay-switch.kicad_sch |       |                   |
| Title: bay-card - bay-switch                      |       |                   |
| Size: A3                                          | Date: | Rev: v0-generated |
| KiCad E.D.A. 9.0.8                                |       | Id: 3/4           |



Fingers: Connector\_PCBEde:BUS\_PCIexpress\_x1 on the card's bottom edge (chamfer the edge, ENIG is fine for a few insertions). Pinout in symgen.SLOT\_PINS.  
SATA P1-P3 (3.3 V) are not connected on purpose: P3 is PWDIS on SATA 3.3 drives and a 3.3 V supply there keeps them from spinning up.